Chunked inference of the LISA Galactic foreground: a step-by-step guide

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Abstract

This note walks through the time–frequency ("chunked") analysis of the Galactic foreground (GF) presented in Buscicchio, Pozzoli et al. (2025), and through the envelope model of Buscicchio, Klein et al. (2025) that it relies on. We follow the pipeline in the order in which it is executed: frequency grid, stationary spectra, orbital modulation, simulated data, coarse graining, Gamma likelihood and sampling. Every equation taken from the two papers is reproduced here, with a link to the paper and its original equation number, so the note can be read on its own. Where our reading differs from the printed equations, we say so explicitly.

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1 The idea

The Galactic foreground is the incoherent superposition of millions of unresolved double white-dwarf binaries in the Milky Way. Its power spectrum is not constant in time. The Galaxy is

concentrated towards the Galactic centre, and LISA's antenna pattern sweeps across it as the constellation orbits the Sun and rotates. The power in each TDI channel therefore rises and falls with a one-year period, twice per year. A process whose statistics vary periodically in time is called *cyclostationary*.

The modulation is *slow*: it changes appreciably over weeks to months. This suggests the following strategy:

1. split the data stream into N_c non-overlapping *chunks* of duration T (one to a few weeks);
2. inside each chunk, treat the data as *stationary*, with a PSD whose galactic part is scaled by the modulation averaged over that chunk;
3. treat different chunks as independent, so the total log-likelihood is a sum over chunks.

The same construction handles *data gaps*. Chunks are simply placed in the stretches of data between gaps, and no special treatment of the gaps is needed.

The total log-likelihood is (Buscicchio, Pozzoli et al. (2025), Eq. (1))

$$\log \mathcal{L}(\boldsymbol{d} \mid \theta) = \sum_{c=1}^{N_c} \log \mathcal{L}^{\mathrm{chunk}}(\boldsymbol{d}^c \mid \theta), \tag{1}$$

where $\boldsymbol{d}^c$ is the data of chunk c and θ the full set of model parameters. The rest of this note explains what goes into a single $\log \mathcal{L}^{\mathrm{chunk}}$.

Notation.

$c = 1, \ldots, N_c$	chunk index
$j \in \{A, E\}$	TDI channel
$k = 1, \ldots, n_f$	fine (full-resolution) frequency bin
$m = 1, \ldots, n_g$	coarse (log-spaced) frequency bin
$l = 1, \ldots, N$	time sample within a chunk
$T,\ \mathrm{d}t$	chunk duration, sampling cadence
$N = T/\mathrm{d}t,\ f_s = 1/\mathrm{d}t$	samples per chunk, sampling frequency
$t_S^c,\ t_E^c$	start and end time of chunk c
$S_{jk}^c(\theta)$	model PSD of channel j, chunk c, frequency f_k

Overview. Figure 1 shows the whole pipeline. There are two branches. The *data* branch (left) is executed *once*, with the injected parameters. The *model* branch (right) is executed at *every* step of the sampler, with the trial parameters θ. The two meet in the likelihood.

2 The frequency grid of a chunk

A chunk of duration T sampled at cadence $\mathrm{d}t$ contains $N = T/\mathrm{d}t$ samples. Its discrete Fourier transform lives on the grid

$$f_k = \frac{k}{T}, \qquad \Delta f = \frac{1}{T}, \tag{2}$$

and n_f denotes the number of grid points inside the analysis band $[f_{\min}, f_{\max}]$, i.e. $n_f \simeq (f_{\max} - f_{\min})\, T$. Longer chunks give finer frequency resolution. With equal-duration chunks, every chunk has the *same* grid.

The Fourier amplitudes are normalised so that their squared modulus is a one-sided PSD estimate (Buscicchio, Pozzoli et al. (2025), Eq. (2)):

$$\tilde{d}_{jk}^c = \sqrt{\frac{2}{W f_s}} \sum_{l=1}^{N} w_l\, d_{jl}^c \exp\left[-\frac{2\pi \imath}{N}\, k\, l\right], \qquad W = \sum_{l=1}^{N} |w_l|^2, \tag{3}$$

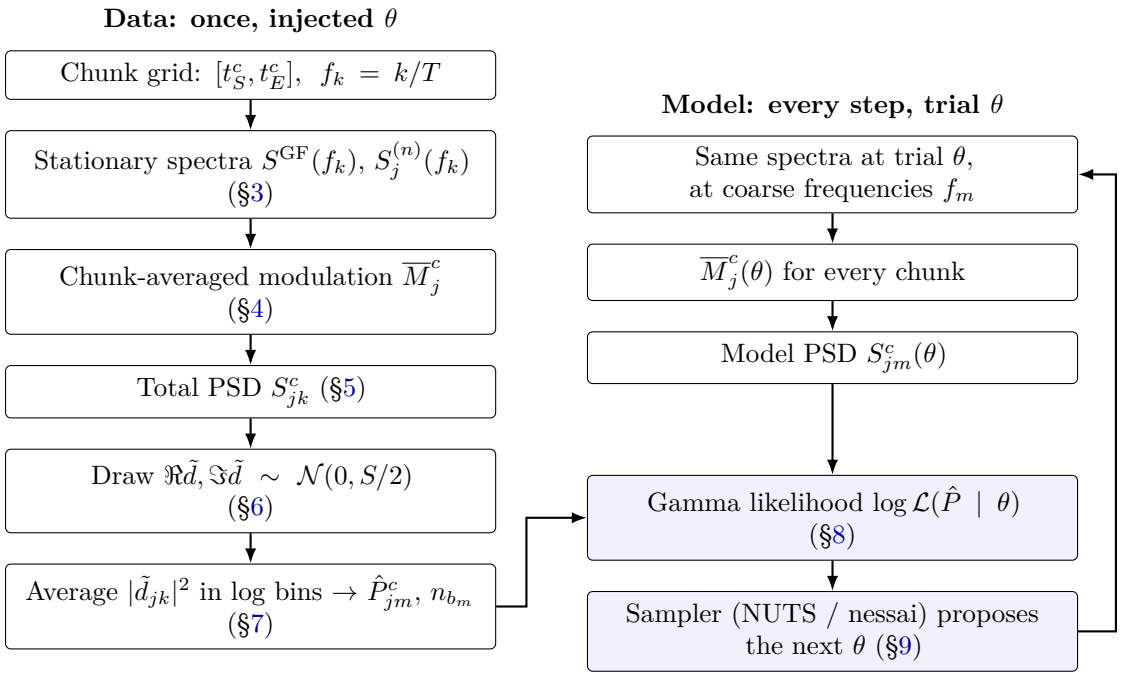


Figure 1: The chunked pipeline. The data are generated once and then frozen. During sampling only the model branch is recomputed: no new random draws and no new coarse graining.

where w_l is a window function. For data generated directly in the frequency domain (Section 6) no window is needed and $W = N$. With this normalisation $\langle|\tilde{d}_{jk}^c|^2\rangle = S_{jk}^c$.

Full-resolution (Whittle) likelihood. If the data in a chunk are Gaussian and stationary, the Fourier amplitudes at different frequencies are independent, and the chunk likelihood is the Whittle likelihood (Buscicchio, Pozzoli et al. (2025), Eq. (3)):

$$\log \mathcal{L}_W^{\rm chunk}(\tilde{\boldsymbol{d}}^c \mid \theta) = -\frac{1}{2} \sum_{j\in\{A,E\}} \sum_{k=1}^{n_f} \left(\frac{|\tilde{d}_{jk}^c|^2}{S_{jk}^c(\theta)} + \log S_{jk}^c(\theta) + \log 2\pi \right). \tag{4}$$

It is exact but expensive. At every sampler step it costs $N_c \times 2 \times n_f$ terms. For example, a two-week chunk has $\Delta f \approx 0.83\,\mu$Hz, so a band of tens of mHz holds tens of thousands of fine bins per channel and per chunk. This motivates the coarse-grained alternative of Section 7.

3 Stationary spectra: Galactic foreground shape and instrument noise

Galactic foreground spectrum. The spectral shape of the GF is the phenomenological model of Karnesis et al. (2021) (Buscicchio, Pozzoli et al. (2025), Eq. (11)):

$$S^{\rm GF}(f) = \frac{A}{2} f^{-7/3} \exp\left[-\left(\frac{f}{f_1}\right)^{\alpha}\right] \left[1 + \tanh\left(\frac{f_{\rm knee} - f}{f_2}\right)\right], \tag{5}$$

with amplitude A, slope parameter α, and characteristic frequencies f_1, $f_{\rm knee}$, f_2. The $f^{-7/3}$ is the inspiral spectrum of a population of circular binaries. The tanh produces the high-frequency drop above $f_{\rm knee}$, where binaries become sparse enough to be resolved and subtracted.

Instrument noise. Two noise sources remain after TDI: test-mass (TM) acceleration noise and optical-metrology-system (OMS) noise (Buscicchio, Pozzoli et al. (2025), Eq. (16) and Buscicchio, Pozzoli et al. (2025), Eq. (17)):

$$S^n_{\rm tm}(f) = \left(A_{\rm n} \times 10^{-15}\,{\rm m\,s^{-2}\,Hz^{-1/2}}\right)^2 \frac{1}{(2\pi c f)^2}\left[1+\left(\frac{0.4\,{\rm mHz}}{f}\right)^2\right]\left[1+\left(\frac{f}{8\,{\rm mHz}}\right)^4\right], \quad (6)$$

$$S^n_{\rm oms}(f) = \left(P \times 10^{-12}\,{\rm m\,Hz^{-1/2}}\right)^2 \left(\frac{2\pi f}{c}\right)^2 \left[1+\left(\frac{2\,{\rm mHz}}{f}\right)^4\right], \quad (7)$$

where c is the speed of light. (The paper calls the TM amplitude A; we write $A_{\rm n}$ to avoid a clash with the GF amplitude.) Both components are projected onto the A and E channels with the TDI transfer functions of Nam et al. (2023), giving the channel noise PSD $S^{(n)}_j(f)$. The injected values are $A_{\rm n} = 2.4$ and $P = 7.9$.

Why these do not depend on the chunk

$S^{\rm GF}(f)$ and $S^{(n)}_j(f)$ depend only on frequency. They are evaluated on the chunk grid f_k, which is the same for every chunk. All the time dependence of the model comes from the modulation factor of Section 4.

Remark: the GF shape evolves with observation time

In Buscicchio, Pozzoli et al. (2025), f_1 and $f_{\rm knee}$ are not constants but functions of the total observation time (Buscicchio, Pozzoli et al. (2025), Eq. (12) and Buscicchio, Pozzoli et al. (2025), Eq. (13)):

$$\log_{10} f_1 = a_1 \log_{10}(T_{\rm obs}) + b_1, \qquad \log_{10} f_{\rm knee} = a_{\rm knee} \log_{10}(T_{\rm obs}) + b_{\rm knee}, \quad (8)$$

with calibrated values $a_1 = -0.25$, $b_1 = -2.7$, $a_{\rm knee} = -0.27$, $b_{\rm knee} = -2.47$. Both slopes are negative. The longer LISA observes, the more binaries are resolved and subtracted, and the cutoff moves to lower frequency.

To mimic a global fit, where the foreground estimate is continuously updated as individual sources are added or removed, the paper simulates the chunks *sequentially*. Each new chunk's spectrum uses the observation time accumulated up to that chunk. The GF spectrum then drifts slowly to lower frequency from chunk to chunk, on top of the up-and-down drift caused by the modulation. If $T_{\rm obs}$ is held fixed, as in the rest of this note, the shape is the same for all chunks.

4 The orbital modulation

The modulation is derived in Buscicchio, Klein et al. (2025) (main text Sec. 6 and Appendix A) in the *low-frequency approximation*. In this limit, A and E behave as two independent effective Michelson interferometers whose orientation follows LISA's orbit.

4.1 Step 1: many sources at a single sky location

Consider a large number of binaries at the same sky position: ecliptic latitude and longitude (b_S, l_S). Their inclinations ι and polarisation angles ψ are random. The squared response is averaged over the gravitational-wave period and over these angles (Buscicchio, Klein et al. (2025), Eq. (27)):

$$\langle h_{A,E}\rangle^2(t, b_S, l_S) = \frac{1}{T}\int_0^T {\rm d}t' \int_{-1}^{1} {\rm d}\cos\iota \int_0^{2\pi} {\rm d}\psi\; h^2_{A,E}. \quad (9)$$

Here T is the *period of the gravitational wave* (minutes to hours). The average removes the fast oscillation of the signal and keeps only its slowly varying amplitude. The result still depends on time t, but only through LISA's annual motion.

The averaged envelope is most conveniently written through the sum and difference of the two channels. Both are short Fourier series in the orbital phase (Buscicchio, Klein et al. (2025), Eq. (A.35)–(A.40)):

$$h_S^2 \equiv \langle h_A \rangle^2 + \langle h_E \rangle^2 = \sum_{n=0}^{4} h_{S;(n)}^2(b_S) \cos\left(n\,\Delta l_L\right), \tag{10}$$

$$h_D^2 \equiv \langle h_A \rangle^2 - \langle h_E \rangle^2 = \sum_{n=0}^{8} h_{D;(n)}^2(b_S) \cos\left(n\,\Delta l_L + 4\bar{l}_S\right), \tag{11}$$

with $\Delta l_L = \Omega t - l_S$, $\bar{l}_S = l_S + \pi/12$, $\Omega = 2\pi/\mathrm{yr}$. The coefficients $h_{S;(n)}^2$, $h_{D;(n)}^2$ are polynomials in $\sin b_S$ and $\cos b_S$ (Buscicchio, Klein et al. (2025), Eq. (A.41)–(A.54)). For example, $h_{S;(0)}^2 = \frac{9}{320}\left(328 + 152\cos^2 b_S - 37\cos^4 b_S\right)$. So the sum $A^2 + E^2$ contains harmonics up to 4Ω, and the difference up to 8Ω.

4.2 Step 2: a model for the sky distribution

The Galaxy is not a point. Its unresolved binaries are modelled with a (deliberately simple, and admittedly unrealistic) bivariate Gaussian on the sky (Buscicchio, Klein et al. (2025), Eq. (A.55)–(A.57)):

$$p(b_S, l_S) = \frac{1}{2\pi\sigma_1\sigma_2} \exp\left[-\frac{1}{2}\,\boldsymbol{\theta}^T \mathbb{M}\,\boldsymbol{\theta}\right], \qquad \boldsymbol{\theta} = \begin{pmatrix} b_S - b_M \\ l_S - l_M \end{pmatrix}, \tag{12}$$

$$\mathbb{M} = \begin{pmatrix} \cos\delta & \sin\delta \\ -\sin\delta & \cos\delta \end{pmatrix} \begin{pmatrix} \sigma_1^{-2} & 0 \\ 0 & \sigma_2^{-2} \end{pmatrix} \begin{pmatrix} \cos\delta & -\sin\delta \\ \sin\delta & \cos\delta \end{pmatrix}. \tag{13}$$

It is centred on the Galactic centre (b_M, l_M), has standard deviations $\sigma_{1,2}$ along its principal axes, and is rotated by an angle δ relative to the ecliptic.

4.3 Step 3: integrate over the sky

The envelope of the whole Galaxy is the single-location envelope weighted by p and integrated over the sky (Buscicchio, Klein et al. (2025), Eq. (A.58)):

$$H_{S,D}^2(t) = \int_{\mathbb{R}} \cos b_S \,\mathrm{d}b_S \int_{\mathbb{R}} \mathrm{d}l_S \; p(b_S, l_S)\, h_{S,D}^2. \tag{14}$$

For simplicity the integral runs over the whole plane rather than over the sphere. This is a good approximation as long as p falls off quickly. The integral can be done in closed form because every term of $h_{S,D}^2$ is a product of sines and cosines of b_S and l_S. The integral of $e^{inb_S} e^{iml_S}$ against a Gaussian is again a Gaussian factor (its characteristic function, Buscicchio, Klein et al. (2025), Eq. (A.59)). Each harmonic is therefore damped by a factor that depends on the widths and orientation of the Galaxy. With the shorthand (Buscicchio, Klein et al. (2025), Eq. (A.63)–(A.69))

$$\sigma_S^2 = \sigma_1^2 + \sigma_2^2, \quad \sigma_D^2 = \sigma_1^2 - \sigma_2^2, \quad \sigma_c^2 = \sigma_D^2 \cos 2\delta, \quad \sigma_s^2 = \sigma_D^2 \sin 2\delta, \quad \sigma_+^2 = \sigma_S^2 + \sigma_c^2, \tag{15}$$

$$\bar{l}_M = l_M + \frac{\pi}{12}, \qquad \Delta\phi_L = \Omega t - l_M, \tag{16}$$

the result is again a finite Fourier series in $\Delta\phi_L$ (Buscicchio, Klein et al. (2025), Eq. (A.61), (A.62)),

$$H_S^2 = \frac{1}{10240}\sum_{n=0}^{4} H_{S;(n)}^2, \qquad H_D^2 = \frac{1}{10240}\sum_{n=0}^{8} H_{D;(n)}^2, \tag{17}$$

where $H_{S;(n)}^2$ and $H_{D;(n)}^2$ contain the n-th harmonic. The first two terms are (Buscicchio, Klein et al. (2025), Eq. (A.70), (A.71))

$$H_{S;(0)}^2 = e^{-\frac{1}{4}\sigma_S^2-\frac{1}{4}\sigma_c^2}\left(120636\cos b_M + 7614\, e^{-2\sigma_+^2}\cos 3b_M - 666\, e^{-6\sigma_+^2}\cos 5b_M\right), \tag{18}$$

$$\begin{aligned} H_{S;(1)}^2 = \sqrt{3}\, e^{-\frac{1}{2}\sigma_S^2}\Big[&-31392\Big(\cosh\tfrac{\sigma_s^2}{2}\sin b_M\cos\Delta\phi_L + \sinh\tfrac{\sigma_s^2}{2}\cos b_M\sin\Delta\phi_L\Big) \\ &- 32112\, e^{-2\sigma_+^2}\Big(\cosh\tfrac{3\sigma_s^2}{2}\sin 3b_M\cos\Delta\phi_L + \sinh\tfrac{3\sigma_s^2}{2}\cos 3b_M\sin\Delta\phi_L\Big) \\ &- 720\, e^{-6\sigma_+^2}\Big(\cosh\tfrac{5\sigma_s^2}{2}\sin 5b_M\cos\Delta\phi_L + \sinh\tfrac{5\sigma_s^2}{2}\cos 5b_M\sin\Delta\phi_L\Big)\Big]. \end{aligned} \tag{19}$$

The remaining terms (Buscicchio, Klein et al. (2025), Eq. (A.72)–(A.83)) follow the same pattern: a damping exponential, times coefficients in b_M, times cos and sin of $n\Delta\phi_L$ (for H_S) or of $n\Delta\phi_L + 4l_M$ (for H_D). Finally, the envelopes of the two channels are (Buscicchio, Klein et al. (2025), Eq. (A.84), (A.85))

$$\boxed{h_{A,0}^2(t) = \tfrac{1}{2}\left(H_S^2 + H_D^2\right), \qquad h_{E,0}^2(t) = \tfrac{1}{2}\left(H_S^2 - H_D^2\right).} \tag{20}$$

These are the squared modulation functions $M_A^2(t)$ and $M_E^2(t)$ used in Buscicchio, Pozzoli et al. (2025). They describe the *power* of the GF in each channel as a function of time. Table 1 lists the parameters under the names used by each source.

Meaning	Klein et al.	Pozzoli et al.	`bahamas`
Galactic-centre latitude	b_M	$\sin\beta$	`lat`
Galactic-centre longitude	l_M	λ	`long`
Variances along the axes	$\sigma_1^2,\ \sigma_2^2$	$\sigma_1^2,\ \sigma_2^2$	`s1`, `s2`
Orientation	δ	$\sin\psi$	`psi` ($=\sin 2\delta$)

Table 1: The five envelope parameters ($\theta_{\mathrm{GF_{II}}}$) as named in Buscicchio, Klein et al. (2025), in Buscicchio, Pozzoli et al. (2025) (which samples the latitude through its sine), and in the code.

A few printed coefficients in App. A of Buscicchio, Klein et al. (2025) contain misprints; the code implements the corrected expressions. Details are in `reports/PAPER_BUGS.md` of the `bahamas` repository.

4.4 Step 4: average over a chunk

Inside a chunk the PSD is assumed stationary, so the modulation is replaced by its average over the chunk (Buscicchio, Pozzoli et al. (2025), Eq. (14)):

$$\overline{M}_j^c = \frac{1}{t_E^c - t_S^c}\int_{t_S^c}^{t_E^c} \mathrm{d}t\ M_j^2(t \mid \theta_{\mathrm{GF_{II}}}). \tag{21}$$

This is a single number per chunk and per channel. Note that this is a *second*, different average. Equation (9) averages over the gravitational-wave period of each binary (fast). Equation (21) averages the resulting slow envelope over the duration of a chunk (weeks).

There is no need to evaluate $M_j^2(t)$ on a fine time grid and integrate it numerically. Equation (20) is a finite sum of terms $\cos(n\Delta\phi_L + \varphi)$ and $\sin(n\Delta\phi_L + \varphi)$, and the average of each term over the chunk is elementary. For $n \geq 1$,

$$\frac{1}{t_E - t_S}\int_{t_S}^{t_E} \mathrm{d}t\, \cos(n\Omega t + \varphi) = \frac{\sin(n\Omega t_E + \varphi) - \sin(n\Omega t_S + \varphi)}{n\,\Omega\,(t_E - t_S)}, \tag{22}$$

and similarly for the sines. Replacing every harmonic by its chunk average gives $\overline{M}_j^c$ *exactly* and at negligible cost. This is what `bahamas` does.

Why the chunk average is a reasonable approximation

The envelope is dominated by low harmonics of the one-year orbit, so it changes little over a week or two. Treating the chunk as stationary with the mean power $\overline{M}_j^c$ discards only the variation of M_j^2 *within* the chunk. The longer the chunk, the larger this error. The shorter the chunk, the fewer frequency bins it has. The chunk duration balances the two.

5 The total PSD of a chunk

Combining the pieces, the PSD of channel j in chunk c is (Buscicchio, Pozzoli et al. (2025), Eq. (10))

$$S_{jk}^c(\theta) = \overline{M}_j^c(\theta_{\mathrm{GF_{II}}})\, S_k^{\mathrm{GF}}(\theta_{\mathrm{GF_I}}) + S_{jk}^{(n)}(\theta_n) + R_{jk}\, S_k^{\mathrm{EF}}(\theta_{\mathrm{EF}}). \tag{23}$$

The first term is the modulated Galactic foreground: a time-independent spectral shape, rescaled chunk by chunk. The second is the stationary instrument noise. The last is an optional stationary, isotropic extragalactic foreground, projected with a chunk-independent response R_{jk}. It is only included when that component is studied, and we set it to zero here. The parameters are grouped as $\theta_{\mathrm{GF_I}} = \{\log_{10} A, \log_{10} f_{\mathrm{knee}}, \log_{10} f_1, \log_{10} f_2\}$ (spectral shape), $\theta_{\mathrm{GF_{II}}}$ (envelope, Table 1) and $\theta_n = \{A_{\mathrm{n}}, P\}$ (noise).

6 Simulating the data

Both the GF and the instrument noise are modelled as zero-mean Gaussian processes. The GF is the sum of very many independent sources, so the central limit theorem applies. The PSD fixes the *variance* at each frequency, not the values of the data. A data set is one random realisation consistent with that variance.

Buscicchio, Pozzoli et al. (2025) generates the data directly in the frequency domain. For each chunk c, channel j and fine frequency f_k, the real and imaginary parts of the Fourier amplitude are drawn independently:

$$\Re\,\tilde{d}_{jk}^c,\ \Im\,\tilde{d}_{jk}^c \ \sim\ \mathcal{N}\Big(0,\ \tfrac{1}{2}S_{jk}^c(\theta_{\mathrm{inj}})\Big)\,, \tag{24}$$

where θ_{inj} are the injected parameters. $\tilde{d}_{jk}^c$ is a *circular* complex Gaussian with zero mean, and its squared modulus has expectation

$$\langle|\tilde{d}_{jk}^c|^2\rangle = \langle(\Re\tilde{d})^2\rangle + \langle(\Im\tilde{d})^2\rangle = S_{jk}^c. \tag{25}$$

The amplitudes themselves average to zero and carry no information. The information sits in their *squared moduli*, whose mean is the PSD. This is why everything that follows works with $|\tilde{d}|^2$.

Since each chunk is drawn independently, the chunks are independent by construction, and no window is needed ($W = N$ in Equation (3)). The data are generated once and then kept fixed for the whole inference.

7 Coarse graining: averaged periodograms

To reduce the cost of Equation (4), the n_f fine frequencies of each chunk are grouped into $n_g \ll n_f$ *logarithmically spaced* intervals. The interval edges are (Buscicchio, Pozzoli et al. (2025), Eq. (5)–(8))

$$\Delta = \frac{\log_{10} f_{\max} - \log_{10} f_{\min}}{n_g}, \tag{26}$$

$$\log_{10} f_m = \log_{10} f_{\min} + m\,\Delta, \tag{27}$$

$$\mathcal{F}_m = \{\, k \mid f_m < f_k < f_{m+1} \,\}, \tag{28}$$

$$n_{b_m} = |\mathcal{F}_m|. \tag{29}$$

Here $\mathcal{F}_m$ is the set of fine frequencies that fall in the m-th interval, and n_{b_m} is how many there are.[1] Within each interval, the squared amplitudes are replaced by their mean, the *averaged periodogram* (Buscicchio, Pozzoli et al. (2025), Eq. (4)):

$$\hat{P}^c_{jm} = \frac{1}{n_{b_m}} \sum_{k \in \mathcal{F}_m} |\tilde{d}^c_{jk}|^2. \tag{30}$$

After this step each chunk and channel is described by n_g numbers $\hat{P}^c_{jm}$ instead of n_f complex amplitudes. Each $\hat{P}^c_{jm}$ is an estimate of the PSD in its interval, whose scatter shrinks as n_{b_m} grows.

Three practical points:

- Because the intervals have equal *logarithmic* width, they are narrow at low frequency and wide at high frequency. n_{b_m} is therefore not n_f/n_g but grows roughly in proportion to f: a few fine bins near $f_{\min}$, hundreds at the high end. Intervals that contain no fine frequency are discarded.
- Averaging assumes that the PSD is nearly constant across an interval. With $n_g = 1000$ (Buscicchio, Pozzoli et al. (2025)) each interval is only a fraction of a percent to about a percent wide in frequency, so this holds wherever the spectrum is smooth.
- The model PSD is evaluated at one representative frequency per interval (in `bahamas`, the midpoint between its first and last fine frequency), giving $S^c_{jm}(\theta)$.

Buscicchio, Pozzoli et al. (2025) uses $f_{\min} = 0.1\,\mathrm{mHz}$ and $n_g = 1000$.[2]

8 The Gamma likelihood

8.1 Why a Gamma distribution

Each $|\tilde{d}_{jk}|^2$ is the sum of the squares of two independent zero-mean Gaussians (real and imaginary part), each with variance $S/2$. That is $\frac{S}{2}\chi^2_2$, an exponential distribution with mean S, which is the special case of the Gamma distribution with shape 1 (cf. Buscicchio, Klein et al. (2025), Eq. (21)). The Gamma family is closed under sums: independent Gamma variables with a common scale add up to a Gamma variable whose shape is the sum of the shapes. The averaged periodogram $\hat{P}^c_{jm}$ is the mean of n_{b_m} uncorrelated squared circular Gaussians, so it is Gamma distributed. (By contrast, the amplitudes $\tilde{d}$ themselves are Gaussian, which is why Equation (4) has a Gaussian form.)

[1] Buscicchio, Pozzoli et al. (2025) prints the third line as $\mathcal{F}_k = \{m \mid f_m < f_k < f_{m+1}\}$. The roles of k and m are swapped there; the version above is the intended one.

[2] The paper quotes $f_{\max} = 2.9\,\mathrm{mHz}$. The `bahamas` template configuration uses $f_{\max} = 0.029\,\mathrm{Hz} = 29\,\mathrm{mHz}$; it is worth confirming which value was used.

Buscicchio, Pozzoli et al. (2025) writes the single-chunk likelihood as (Buscicchio, Pozzoli et al. (2025), Eq. (9))

$$\log \mathcal{L}_G^{\text{chunk}}(\hat{P}^c \mid \theta) = -\sum_{j\in\{A,E\}} \sum_{m=1}^{n_g} \left[\log\Gamma\left(\frac{n_{b_m}}{2}\right) + \frac{n_{b_m}}{2} \log S_{jm}^c(\theta) - \left(\frac{n_{b_m}}{2} - 1\right) \log \hat{P}_{jm}^c + \frac{n_{b_m}}{2} \frac{\hat{P}_{jm}^c}{S_{jm}^c(\theta)} \right]. \tag{31}$$

The two roles are worth stating clearly:

- $\hat{P}_{jm}^c$ (with a hat) is the **data**: the averaged periodogram of Equation (30), computed once from the simulated amplitudes.
- $S_{jm}^c(\theta)$ is the **model**: the PSD of Equation (23) evaluated at whatever θ the sampler is currently trying.

The injected ("true") PSD does not appear in the likelihood at all. It was only used to generate the data in Section 6. When $\theta = \theta_{\text{inj}}$, the model coincides with it.

8.2 Comparison with the textbook Gamma likelihood

The Gamma density with shape $k > 0$ and scale $\vartheta > 0$ is (NIST/SEMATECH e-Handbook, Sec. 1.3.6.6.11; Casella & Berger, *Statistical Inference*, Sec. 3.3)

$$p(x \mid k, \vartheta) = \frac{x^{k-1}\, e^{-x/\vartheta}}{\Gamma(k)\, \vartheta^k}, \qquad \langle x \rangle = k\vartheta, \qquad \mathrm{Var}(x) = k\vartheta^2. \tag{32}$$

For n independent samples with the *same* k and ϑ, the log-likelihood is

$$\ell(k, \vartheta) = (k-1) \sum_{i=1}^{n} \log x_i - \frac{1}{\vartheta} \sum_{i=1}^{n} x_i - n \log\Gamma(k) - nk \log \vartheta. \tag{33}$$

In Equation (31), every pair (j, m) is a *single* observation with its own shape and scale. The sum over channels and bins plays the role of the sum over samples in Equation (33), with $n = 1$ for each term. Writing one term of Equation (33) with the overall minus sign of Equation (31) pulled out:

Textbook, one sample	Eq. (31), one (j,m)	Identification
x	$\hat{P}_{jm}^c$	the data
$(k-1)\log x$	$+\left(\frac{n_{b_m}}{2} - 1\right)\log \hat{P}$	shape $k = n_{b_m}/2$
$-\log\Gamma(k)$	$-\log\Gamma\left(\frac{n_{b_m}}{2}\right)$	same k
$-x/\vartheta$	$-\frac{n_{b_m}}{2}\, \hat{P}/S$	scale $\vartheta = 2S/n_{b_m}$
$-k\log\vartheta$	$-\frac{n_{b_m}}{2}\log S$	$= -k\log\vartheta$ up to $-\frac{n_{b_m}}{2}\log\frac{2}{n_{b_m}}$

The last entry differs only by a term that depends on n_{b_m} alone. That term does not depend on θ, so Buscicchio, Pozzoli et al. (2025) drops it. In summary,

$$\hat{P}_{jm}^c \sim \text{Gamma}\left(k = \frac{n_{b_m}}{2}, \quad \vartheta = \frac{2S_{jm}^c(\theta)}{n_{b_m}}\right), \qquad \langle \hat{P} \rangle = k\vartheta = S_{jm}^c(\theta). \tag{34}$$

Interpretation.

- *The model PSD is the mean, and it sets the scale.* S is not the scale parameter itself but the mean $k\vartheta$. Since $\vartheta = 2S/n_{b_m}$ is proportional to S, changing S at fixed n_{b_m} simply stretches the distribution of $\hat{P}$ without changing its shape.
- *The number of averaged bins sets the shape.* The relative scatter is $\sqrt{\mathrm{Var}}/\langle\hat{P}\rangle = 1/\sqrt{k}$. For small n_{b_m} the distribution is skewed (an exponential for $k = 1$). For large n_{b_m} it approaches a narrow Gaussian centred on S. High-frequency intervals, which average many bins, therefore constrain the model more tightly.
- *What the sampler sees.* Only two terms depend on θ:

$$\log \mathcal{L}_G^{\text{chunk}} = -\sum_{j,m} \frac{n_{b_m}}{2} \left[\log S_{jm}^c(\theta) + \frac{\hat{P}_{jm}^c}{S_{jm}^c(\theta)} \right] + \text{const.} \tag{35}$$

  Each bracket is minimised at $S = \hat{P}$, and each is weighted by n_{b_m}. For $n_{b_m} = 1$ this reduces to the θ-dependent part of the Whittle likelihood, Equation (4). So the Gamma likelihood is the natural generalisation of Whittle to averaged periodograms.

Remark: counting degrees of freedom

A complex Fourier amplitude carries *two* real Gaussian degrees of freedom, so each $|\tilde{d}_{jk}|^2$ is Gamma with shape 1 (as in Buscicchio, Klein et al. (2025), Eq. (21)). The mean of n_{b_m} of them is therefore Gamma with shape n_{b_m} and scale S/n_{b_m}, with variance S^2/n_{b_m}. Equation (31) instead uses shape $n_{b_m}/2$, which corresponds to variance $2S^2/n_{b_m}$. The θ-dependent part of Equation (31) is then exactly one half of the exact one:

$$\mathcal{L}_{\text{Eq. (9)}}(\theta) \propto \mathcal{L}_{\text{exact}}(\theta)^{1/2}.$$

This is equivalent to tempering the likelihood with inverse temperature 1/2. The location of the posterior is unchanged, but its widths are inflated by about $\sqrt{2}$. The Whittle likelihood of Equation (4) carries the same factor $\frac{1}{2}$ per complex amplitude, so the two likelihoods remain consistent with each other. Doubling n_{b_m} in Equation (31), and dropping the $\frac{1}{2}$ in Equation (4), gives the exact versions. We flag this as a point to discuss with the authors.

9 Inference

Putting Equation (1) and Equation (31) together gives $\log \mathcal{L}(\hat{\boldsymbol{P}} \mid \theta)$. Bayes' theorem then gives the posterior $p(\theta \mid \hat{\boldsymbol{P}}) \propto \mathcal{L}(\hat{\boldsymbol{P}} \mid \theta)\, \pi(\theta)$. In Buscicchio, Pozzoli et al. (2025) the parameter vector has 14 components: 4 for the GF spectrum ($\theta_{\mathrm{GF_I}}$), 5 for the envelope ($\theta_{\mathrm{GF_{II}}}$), 3 for the extragalactic foreground (θ_{EF}), and the two noise amplitudes A_{n} and P. All have uniform priors. Without the extragalactic foreground there are 11.

At every step the sampler proposes a trial θ, and only the model branch of Figure 1 is recomputed: $\overline{M}_j^c(\theta)$ for every chunk, the spectra at the coarse frequencies, and the model PSD $S_{jm}^c(\theta)$. The data $\hat{P}_{jm}^c$ never change. Because every ingredient is analytic (the envelope of Equation (20), its chunk average and the spectra), the whole chain from θ to $\log \mathcal{L}$ is cheap and differentiable. Two kinds of sampler are used:

- **Hamiltonian Monte Carlo** with the No-U-Turn Sampler (NUTS, Hoffman & Gelman 2014), as implemented in NumPyro. HMC moves through parameter space along trajectories guided by the gradient $\nabla_\theta \log p(\theta \mid \hat{\boldsymbol{P}})$, which is obtained by automatic (reverse-mode) differentiation in `jax`. No gradient has to be derived by hand. This makes HMC efficient even with many correlated parameters.

- **Nested sampling** with `nessai`, which uses normalising flows to propose new live points. Nested sampling is not an MCMC method. It explores the prior volume shell by shell and returns the Bayesian evidence $\mathcal{Z} = \int \mathcal{L}\,\pi\,\mathrm{d}\theta$ together with the posterior. The evidence is what allows the modulated (quasi-stationary) model to be compared with a stationary one, in which $\overline{M}^c_j$ is constant, via their Bayes factor.

Buscicchio, Pozzoli et al. (2025) finds the same posteriors with both samplers and with both likelihoods (Whittle and Gamma).

10 Summary

The chunked pipeline in nine steps

1. Split the observation into N_c chunks $[t^c_S, t^c_E]$ of duration T; each has the frequency grid $f_k = k/T$.
2. Evaluate the stationary spectra $S^{\mathrm{GF}}(f_k)$ and $S^{(n)}_j(f_k)$ (Eqs. 5–7); they are the same for every chunk.
3. For each chunk and channel, compute the chunk-averaged modulation $\overline{M}^c_j$ (Equation (21)) analytically from the sky-integrated envelope (Equation (20)).
4. Form the total PSD $S^c_{jk} = \overline{M}^c_j S^{\mathrm{GF}}_k + S^{(n)}_{jk}$ (Equation (23)).
5. Draw $\Re\tilde{d}^c_{jk}$ and $\Im\tilde{d}^c_{jk}$ independently from $\mathcal{N}(0, S^c_{jk}/2)$ with the injected parameters (Equation (24)).
6. Group the fine frequencies into n_g log-spaced intervals, average $|\tilde{d}^c_{jk}|^2$ in each, and keep $\hat{P}^c_{jm}$ and n_{b_m} (Equation (30)).
7. For a trial θ, recompute $\overline{M}^c_j(\theta)$ and $S^c_{jm}(\theta)$ at the coarse frequencies.
8. Evaluate the Gamma likelihood (Equation (31)) and sum it over chunks (Equation (1)).
9. Let the sampler (NUTS or nessai) propose the next θ and return to step 7.

Steps 1–6 are done once. Steps 7–9 are repeated until the posterior is sampled.

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